

mPLUG-Octopus: The Versatile Assistant Empowered by A Modularized End-to-End Multimodal LLM

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ABSTRACT

Inspired by the recent developments of large language models (LLMs), we propose mPLUG-Octopus, a versatile conversational assistant designed to provide users with coherent, engaging, and helpful interaction experiences in both text-only and multi-modal scenarios. Unlike traditional pipeline chatting systems, mPLUG-Octopus offers a diverse range of creative capabilities including open-domain QA, multi-turn chatting, and multi-modal creation, all built with a unified multimodal LLM without relying on any external API. With the modularized end-to-end multimodal LLM technology, mPLUG-Octopus efficiently facilitates engaging and open-domain conversation experience. It exhibits a wide range of uni/multi-modal elemental capabilities, enabling it to seamlessly communicate with users on open-domain topics and engage in multi-turn conversations. It also assists users in accomplishing various content creation and application tasks. Our conversational assistant can also be deployed on smart hardware to drive advanced AIGC applications.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

multi-modal dialogue chatbot, modularized model, multi-modal model, large language model, instruction tuning

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1 INTRODUCTION

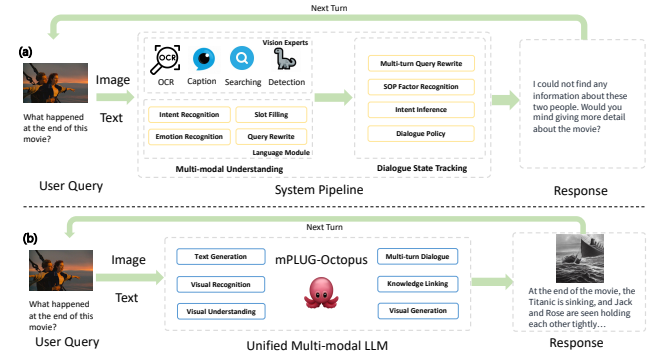


Figure 1: Comparison between traditional dialogue system and our proposed modularized end-to-end mPLUG-Octopus.

Building an intelligent versatile conversational assistant that can converse like humans has been a long-term goal towards Artificial General Intelligence(AGI). The assistant should be able to generate coherent and engaging responses in an open and multi-turn conversation scenario while also possessing various skills to assist humans in their work. Large language models [1, 3, 7] (LLMs) have pushed the boundaries of open-domain text-based dialogue systems by scaling up model size and training corpora. However, it is challenging to support both uni-modal and multi-modal formats in an open-domain and multi-turn conversation scenario. Previous dialogue systems [5, 6, 8, 9, 11] relied on complex modules such as

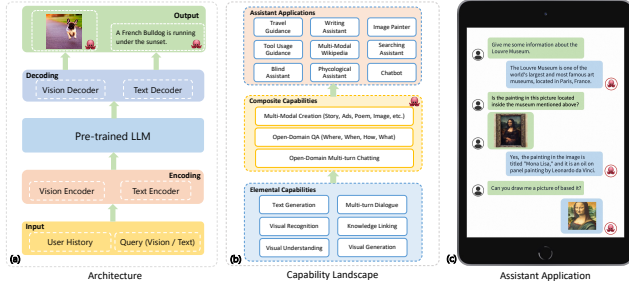


Figure 2: The overview of mPLUG-Octopus assistant and the application. (a) The architecture of mPLUG-Octopus and the workflow. (b) The capability landscape of mPLUG-Octopus and the application. (c) The demo illustration of our proposed mPLUG-Octopus multi-modal chatbot.

visual understanding, dialogue state tracking, text generation, visual generation, etc. These complex modules depended on external APIs, resulting in high deployment costs and cascading errors.

Building upon our multimodal LLM mPLUG-Owl [10], we present mPLUG-Octopus, a versatile conversational assistant that enhances its dialogue and assistant capabilities, including open-domain question answering, multi-turn conversation, and multimodal content creation. Different from traditional dialogue systems, our conversational assistant can provide a faster and smoother conversation experience because it is based on one unified end-to-end model, rather than multiple complex modules and external APIs. It can also deliver a more diverse and coherent multi-modal and multi-turn conversation experience as it is empowered by multimodal LLM. Furthermore, our conversational assistant possesses rich elemental capabilities in both single-modal and multi-modal understanding and generation as in Figure 2, helping users efficiently complete tasks through dialogue. For instance, imagine that we are planning to visit Paris for a tourism, we can ask mPLUG-Octopus to help us introduce Paris and collect detailed information for famous attractions in Paris. It can also help to arrange our itinerary and recommend us the famous football players here while we would like to watch a football game. Finally, we can even generate a multi-modal tweet about the football star to conclude this enjoyable journey.

2 MPLUG-OCTOPUS

2.1 Model Overview

Large language models (LLMs) exhibit substantial zero-shot generalization capabilities, demonstrating their proficiency in accomplishing tasks that have not been previously encountered. To this end, we propose mPLUG-Octopus, an end-to-end multi-modal assistant that employs the powerful LLM for unified understanding and generation. As depicted in Figure 2, mPLUG-Octopus incorporates a visual encoder and a text encoder designed to encode different modalities derived from dialogue history $\mathcal{H}$ and the current user query Q . Subsequently, a pre-trained LLM is employed for dialogue history comprehension and query interpretation, resulting in multi-modal response representations. This representation can then be decoded by either an image decoder or a text decoder to generate response.

To harness the inherent capabilities of the pre-trained LLM, we train mPLUG-Octopus through a two-stage approach. The initial

stage aims to align the visual representation of the visual encoder with the pre-trained LLM by generating captions for given images using a vast array of image-text pairs. Then, we employ the low-rank adaption [2] with a few trainable parameters in the frozen LLM by instruction tuning. Moreover, we require the model to learn when it should produce an image response rather than a text response. Therefore, we add a special [IMG] token to the vocabulary of the LLM. If the model generates the [IMG] token, it will feed the output feature into the image decoder which is a frozen stable diffusion model [4] for yielding image output. Through two-stage training, mPLUG-Octopus is capable of generating responses with different types of modalities.

2.2 Capability Landscape and Applications

mPLUG-Octopus is structured with a whole capability landscape encompassing elemental capabilities, composite capabilities, and assistant applications. Specifically, the modularized design and two-stage training empower mPLUG-Octopus to acquire various elemental capabilities, such as text generation, visual recognition, and visual generation, etc. For instance, the LLM in mPLUG-Octopus possesses the elemental capability of text generation and multi-turn dialogue, and the visual encoder equips mPLUG-Octopus with the ability to recognize the images.

Built on these elemental capabilities, mPLUG-Octopus integrates them into several sophisticated composite capabilities, encompassing multi-modal creation, open-domain question-answering, and open-domain multi-turn conversation. Specifically, armed with the elemental capabilities of text generation, visual understanding, and visual generation, mPLUG-Octopus is adept at initiating the production of creative content within a given context. Moreover, mPLUG-Octopus can perform open-domain QA based on user input by amalgamating elemental capabilities such as visual understanding and knowledge linking, facilitated by the LLM. As illustrated in Figure 2, endowed with the wide range of capabilities, mPLUG-Octopus can support numerous potential application scenarios and impart a substantial impact on real-world situations.

2.3 Online Demo

We deploy our demo on a single machine with an NVIDIA-V100 GPU. The demo features an interface that enables users to chat with mPLUG-Octopus via a text input box and to upload images or videos. The chat history is displayed on the web page for easy reference. To expedite inference, we implemented the entire model with 8-bit quantization techniques. Additionally, we enhanced the user experience by implementing stream decoding of the output, which reduces waiting times during interaction.

3 CONCLUSION

We propose a versatile conversational assistant, mPLUG-Octopus, which provides a wide range of creative capabilities based on a unified multimodal LLM that does not rely on external APIs. With its modularized end-to-end multimodal LLM, mPLUG-Octopus can deliver engaging and open-domain conversation experiences efficiently. Our versatile assistant can seamlessly communicate with users on a variety of topics and engage in both text-only and multi-modal conversations with multi-turn interaction.

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